

Stakeholder

Who pays / benefits · buy/build · latency day · risks

Product one-liner

UMI is the open GoPro gripper → Diffusion Policy pipe that buys action-diverse in-the-wild demos without wheeling a robot into the cafe.

Who pays / who benefits

Consumers vs non-consumers

Pays / benefits

- Labs wanting wild visual diversity without Franka fleets
- Companies shipping policies across multiple arms
- Course / BEHAVIOR teams needing handheld BC recipe
- Anyone needing toss/fold/wash demos teleop can't collect

Non-consumers

- No SLAM / VIO expertise on staff
- Need dexterous hands (parallel-jaw ceiling)
- Pure language-VLA without wrist cams
- Need 76k-scale robot-in-building traj → DROID

Buy / build decision tree

UMI vs DROID vs ALOHA

1. IF Dynamic + bimanual BC from humans

→ Build UMI (~\$400 + printer) before teleop carts

2. IF 76k Franka traj in real buildings

→ DROID-style robot-in-wild still wins scale

3. IF Fine bimanual lab skills

→ ALOHA/ACT joint teleop still strong

4. IF Any toss-class / dynamic deploy

→ ALWAYS budget latency calibration day

Contrast poles — UMI / DROID / ALOHA

Who goes into the wild?

UMI

Human + gripper
in the wild

No robot at collect
~\$371 · relative EE

DROID

Franka robot
in the wild

76k / 564 scenes
Quest teleop

ALOHA

Leader-follower
in the lab

Joint-space
embodiment-specific

Always budget a latency calibration day

Not optional for toss-class tasks

87.5%

with matching

57.5%

without · -30 pp

App. A

QR + cross-corr recipe

- Measure cam / proprio / gripper / arm — examples 100/120 ms are NOT constants.
- Sync to highest-latency stream; drop stale actions; send-ahead by exec latency.
- Skip this day → jittery motion; grasp may limp along; toss fails.
- Stakeholder rule: first deploy schedule includes calibration before demos.

Cost & throughput

BOM 73 + 298 → 3× spacemouse enables toss demos teleop cannot

73 + 298

printed + GoPro

780 g

handheld mass

~48%

of hand speed (cup)

~64%

of hand speed (toss)

>3×

vs spacemouse teleop

0

toss demos via spacemouse in 15 min

Fig. 11 — throughput

UMI vs bare hand vs spacemouse · 15 min same operator



UMI enables dynamic toss demos that spacemouse could not collect

Risks for stakeholders

SLAM support · sunlight · jaw ceiling

SLAM support burden ORB-SLAM3 fork + Docker; textureless rooms fail; need mapping discipline.

Sunlight / domain shift Known wild-cup failure under direct sun — indoor/outdoor mix needs care.

Parallel-jaw ceiling No dex hand path; contact-rich kitchen may need different interface.

Operator-eval inflation Success rubrics subjective; demand videos + matched inits in contracts.

Kin-filter data drop Some demos silently discarded per embodiment — budget recollect.

Motion for stakeholders

$T_a=6$ @10Hz · relative EE · latency = product feature

$T_a=6$

@10 Hz (20 Hz toss)

Relative SE(3)

not abs / not delta

Latency match

first-class for dynamic

Wrist-only

155° fisheye obs

Buyer ask

Does the vendor ship latency calibration + relative action interface, or only a gripper STL? UMI's product is the INTERFACE, not the plastic.

Capability headline card

What you can buy with the interface

Cup lab	100% UR5 / 90% Franka	305 demos
Toss	87.5% (no lat 57.5%)	280 demos
Cloth fold	70% (no inter-grip 30%)	250 demos
Dish wash	70% (ResNet 0%)	258 demos
Cup wild	71.7% vs narrow 0%	1400 / 30 loc

Numbers tattoo — pin these

Every deck carries the same spine numbers

100% / 90%

cup UR5 / Franka

87.5%

toss w/ latency

57.5%

toss no latency

71.7%

wild 43/60 OOD

0%

narrow → wild OOD

100% vs 25%

rel vs abs action

55%

no fisheye (69°)

70% / 30%

cloth ± inter-grip

70% / 0%

dish ViT / ResNet

1400 / 30

wild demos / locs

Ta=6

@10Hz (20Hz toss)

73 + 298

BOM · 780g

Exam bite — Stakeholder

Buy UMI when you need action-diverse wild demos without robot logistics; buy DROID for Franka-scale scene hours; buy ALOHA for lab bimanual joints. Always schedule latency calibration. Risks: SLAM, sunlight, jaw ceiling.

Stakeholder — close

Takeaways + exam bite

1. Product: GoPro gripper → DP pipe; action-diverse wild demos without robot-in-cafe.
2. Decision tree: UMI / DROID / ALOHA by goal; always latency day.
3. Risks: SLAM support, sunlight, parallel-jaw ceiling, eval inflation.
4. Motion is a product feature: $T_a=6$, relative SE(3), latency matching.

EXAM BITE

UMI = human logistics not robot logistics; budget latency day; vs DROID/ALOHA by goal.